

Course: Database Management Systems Lab

Project: Employee Management System

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INTRODUCTION

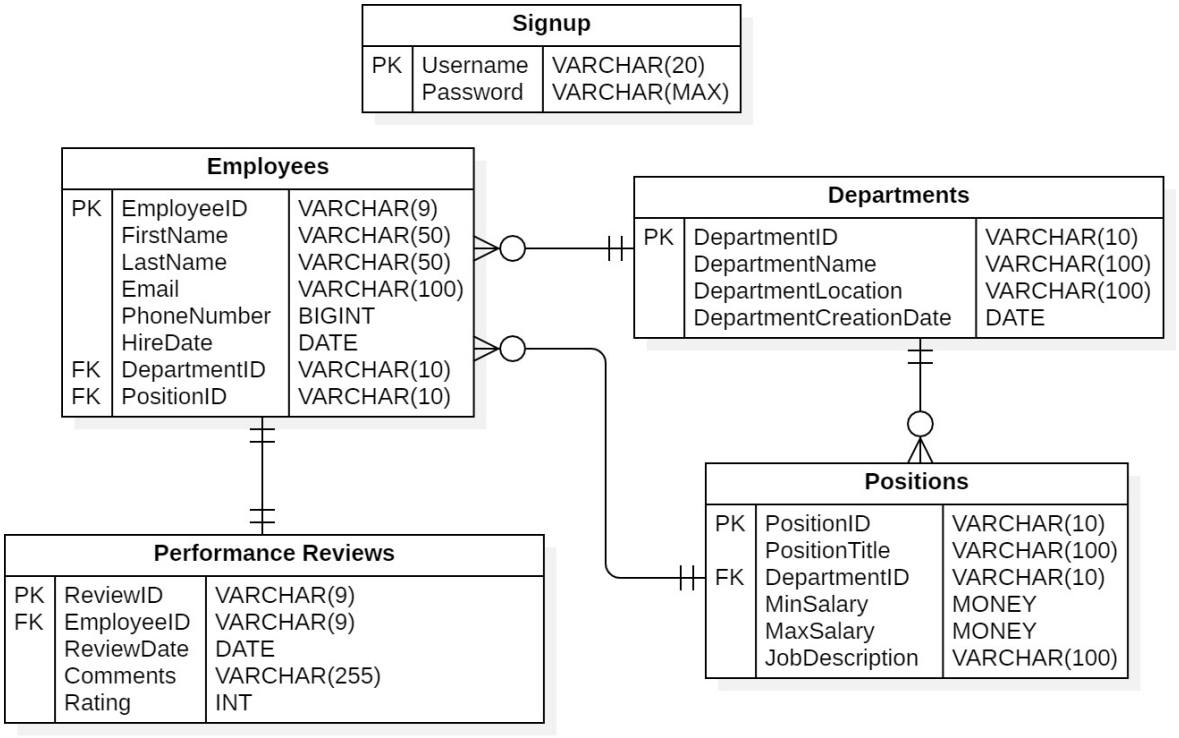
This report details the implementation of an Employee Management System (EMS) using SQL Server for database management and Python for application logic. The EMS is designed to efficiently handle various aspects of employee information, including departments, positions, salaries, and user authentication.

An Employee Management System (EMS) is a comprehensive software solution designed to manage and streamline various aspects of employee administration within an organization. This project, developed using SQL and Python, leverages the robust data management capabilities of SQL and the versatility of Python to create a user-friendly, efficient, and scalable system.

The EMS project aims to facilitate the automation of key functions such as employee record keeping, payroll management, and record keeping for departments and positions within an organization. The SQL database serves as the backbone of the system, storing all data in a structured format, while Python scripts handle the application logic, data processing, and user's interactions with the system and database, and interface elements.

By integrating SQL and Python, the system ensures data integrity, security, and quick access to information, making it an invaluable tool for organizations looking to enhance productivity and accuracy in managing their workforce.

ER DIAGRAM




```
1) One Department Can Have Many Positions In It.
2) One Position Can Be Common Among Several Employees.
3) Many Employees Can Work Inside A Department.
4) One Employee Can Have Many Performance Reviews.
5) Sign-Up Does Not Have Any Relationship With Other Tables.
```

SQL SERVER IMPLEMENTATION

1) **Creating Database:** The EMS begins with the creation of a dedicated database named 'EmployeeManagementDB'. This database is the foundation where all data related to employees, departments, positions, salaries, and user credentials are stored.

2) **Creating Tables:** Several tables are created within the 'EmployeeManagementDB' database to organize and store data:

   - **Departments Table:** This table holds information about different departments within the organization. Each department has a unique identifier and a name.
   - **Positions Table:** This table stores details about various job positions. Each position is associated with a specific department and has a unique identifier and a name.
   - **Employees Table:** This table contains employee data, including personal details, position, department, and hire date. Each employee has a unique identifier.
   - **Performance Reviews Table:** This table records information regarding the performance of employees, including the comments and performance rating. Each review is linked to an employee by foreign key constraint.
   - **Signup Table:** This table manages user authentication details, storing usernames, password hashes, and salts for secure login processes.

3) **Creating Database View:** A database view named 'EmployeeManagementView' is created to provide a comprehensive view of employee information by joining the 'Employees', 'Departments', 'Positions' and 'Salaries' tables. This view simplifies querying and reporting by presenting a consolidated dataset.

4) **Inserting Data Into The Tables:** Data insertion involves populating the tables with initial data. For example, departments are added to the 'Departments' table, positions are added to the 'Positions' table, employee records are added to the 'Employees' table and performance review records for employees are added to the 'Performance Reviews' table. This step is crucial for setting up the EMS with relevant organizational data.

5) **Displaying Data From The Tables:** Data retrieval operations are performed to display information stored in the tables. For instance, querying the 'Employees' table can provide a list that contains details of all employees along with their department, and position IDs. This feature is essential for generating reports and accessing employee data as needed.

6) **Error And Exception Handling Mechanisms:** Robust error and exception handling mechanisms are implemented to ensure the EMS can gracefully handle any unexpected issues during database operations. This includes using try-catch blocks to manage SQL errors and providing meaningful error messages and error codes to users.

7) **Batch Separation:** Batch separation involves organizing SQL statements into batches using the ‘GO’ statement. This practice helps manage large scripts by dividing them into smaller, executable segments, improving readability and maintainability. The involvement of this feature will also ensure that if any query fails during execution, all other queries before and after that query do execute successfully.

SQL QUERY SCREENSHOTS

QUERY 01 & 02:

```
-- CHECK AND CREATE DATABASE
BEGIN TRY
    IF NOT EXISTS (SELECT * FROM SYS.DATABASES WHERE name = 'EmployeeManagementDB')
        BEGIN
            CREATE DATABASE EmployeeManagementDB;
        END;
END TRY

BEGIN CATCH
    PRINT CONCAT('ERROR OCCURED WHILE CREATING THE DATABASE: ', ERROR_MESSAGE());
    PRINT CONCAT('ERROR CODE: ', CAST(@@ERROR AS NVARCHAR(10)));
END CATCH;
GO
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USE EmployeeManagementDB;
GO
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```

QUERY 03:

```
-- DEPARTMENTS TABLE
BEGIN TRY
    IF NOT EXISTS (SELECT * FROM EmployeeManagementDB.SYS.TABLES WHERE NAME = 'Departments')
        BEGIN
            CREATE TABLE Departments (
                DepartmentID VARCHAR(10) PRIMARY KEY,
                DepartmentName VARCHAR(100) NOT NULL,
                DepartmentLocation VARCHAR(100) NOT NULL,
                CreationDate DATE NOT NULL DEFAULT CURRENT_TIMESTAMP,
                CONSTRAINT Check_DepartmentID CHECK (DepartmentID LIKE 'DEPT[0-9][0-9][0-9][0-9][0-9][0-9]')
            );
        END;
END TRY

BEGIN CATCH
    PRINT CONCAT('ERROR OCCURED WHILE CREATING THE TABLE ''DEPARTMENTS'': ', ERROR_MESSAGE());
    PRINT CONCAT('ERROR CODE: ', CAST(@@ERROR AS NVARCHAR(10)));
END CATCH;
GO
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```

QUERY 04:

```
-- POSITIONS TABLE
BEGIN TRY
    IF NOT EXISTS (SELECT * FROM EmployeeManagementDB.SYS.TABLES WHERE NAME = 'Positions')
        BEGIN
            CREATE TABLE Positions (
                PositionID VARCHAR(10) PRIMARY KEY,
                PositionTitle VARCHAR(100) NOT NULL,
                DepartmentID VARCHAR(10) FOREIGN KEY (DepartmentID) REFERENCES Departments(DepartmentID),
                MinSalary MONEY DEFAULT NULL,
                MaxSalary MONEY DEFAULT NULL,
                JobDescription VARCHAR(100) DEFAULT NULL,
                CONSTRAINT Check_PositionID CHECK (PositionID LIKE 'POST[0-9][0-9][0-9][0-9][0-9][0-9]')
            );
        END;
END TRY

BEGIN CATCH
    PRINT CONCAT('ERROR OCCURED WHILE CREATING THE TABLE ''POSITIONS'': ', ERROR_MESSAGE());
    PRINT CONCAT('ERROR CODE: ', CAST(@@ERROR AS NVARCHAR(10)));
END CATCH;
GO
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```

QUERY 05:

```
-- EMPLOYEES TABLE
BEGIN TRY
    IF NOT EXISTS (SELECT * FROM EmployeeManagementDB.SYS.TABLES WHERE NAME = 'Employees')
        BEGIN
            CREATE TABLE Employees (
                EmployeeID VARCHAR(9) PRIMARY KEY,
                FirstName VARCHAR(50) NOT NULL,
                LastName VARCHAR(50) NOT NULL,
                Email VARCHAR(100) NULL,
                PhoneNumber BIGINT NULL,
                HireDate DATE NOT NULL,
                BaseSalary MONEY NOT NULL,
                Bonus MONEY NOT NULL,
                AnnualCompensation AS (BaseSalary + Bonus) * 12,
                PayDate DATE NOT NULL,
                DepartmentID VARCHAR(10) FOREIGN KEY (DepartmentID) REFERENCES Departments(DepartmentID),
                PositionID VARCHAR(10) FOREIGN KEY (PositionID) REFERENCES Positions(PositionID),
                CONSTRAINT Check_EmployeeID CHECK (EmployeeID LIKE 'EMP[0-9][0-9][0-9][0-9][0-9]'),
                CONSTRAINT Check_PhoneNumber CHECK (PhoneNumber LIKE '[0-9][0-9][0-9][0-9][0-9][0-9][0-9][0-9][0-9][0-9]'),
                CONSTRAINT Check_Email CHECK (Email LIKE '%@%.%')
            );
        END;
END TRY

BEGIN CATCH
    PRINT CONCAT('ERROR OCCURED WHILE CREATING THE TABLE ''EMPLOYEES'': ', ERROR_MESSAGE());
    PRINT CONCAT('ERROR CODE: ', CAST(@@ERROR AS NVARCHAR(10)));
END CATCH;
GO
-- FA22-BESE-0003 - FA22-BESE-0007 - FA22-BESE-0029
```

QUERY 06:

```
-- PERFORMANCE REVIEW TABLE
BEGIN TRY
    IF NOT EXISTS (SELECT * FROM EmployeeManagementDB.SYS.TABLES WHERE NAME = 'PerformanceReview')
        BEGIN
            CREATE TABLE PerformanceReview (
                ReviewID VARCHAR(9) PRIMARY KEY,
                EmployeeID VARCHAR(9) FOREIGN KEY (EmployeeID) REFERENCES Employees(EmployeeID),
                ReviewDate DATE DEFAULT GETDATE(),
                Comments VARCHAR(255),
                Rating INT,
                CONSTRAINT Check_Rating CHECK (Rating LIKE '[0-9]'),
                CONSTRAINT Check_ReviewID CHECK (ReviewID LIKE 'REV[0-9][0-9][0-9][0-9][0-9][0-9]')
            );
        END;
END TRY

BEGIN CATCH
    PRINT CONCAT('ERROR OCCURED WHILE CREATING THE TABLE ''PERFORMANCE REVIEW'': ', ERROR_MESSAGE());
    PRINT CONCAT('ERROR CODE: ', CAST(@@ERROR AS NVARCHAR(10)));
END CATCH;
GO
-- FA22-BESE-0003 - FA22-BESE-0007 - FA22-BESE-0029
```

QUERY 07:

```
-- USER CREDENTIALS TABLE
BEGIN TRY
    IF NOT EXISTS (SELECT * FROM EmployeeManagementDB.SYS.TABLES WHERE NAME = 'Sign_Up')
        BEGIN
            CREATE TABLE Sign_Up (
                Username VARCHAR(20) PRIMARY KEY,
                Password VARCHAR(MAX)
            );
        END;
END TRY

BEGIN CATCH
    PRINT CONCAT('ERROR OCCURED WHILE CREATING THE TABLE ''Sign_Up'': ', ERROR_MESSAGE());
    PRINT CONCAT('ERROR CODE: ', CAST(@@ERROR AS NVARCHAR(10)));
END CATCH;
GO
-- FA22-BESE-0003 - FA22-BESE-0007 - FA22-BESE-0029
```

QUERY 08:

```
-- DATABASE VIEW
BEGIN TRY
    IF NOT EXISTS (SELECT * FROM EmployeeManagementDB.SYS.VIEWS WHERE OBJECT_ID = OBJECT_ID('EmployeeManagementView'))
        BEGIN
            EXEC('
                CREATE VIEW EmployeeManagementView AS
                    SELECT
                        EMP.EmployeeID, EMP.FirstName, EMP.LastName, EMP.Email, EMP.PhoneNumber, EMP.HireDate, EMP.BaseSalary,
                        EMP.Bonus, EMP.AnnualCompensation, EMP.PayDate, DP.DepartmentID, DP.DepartmentName, DP.DepartmentLocation,
                        DP.CreationDate AS DepartmentCreationDate, POST.PositionID, POST.PositionTitle, POST.MinSalary, POST.MaxSalary,
                        POST.JobDescription, REV.ReviewID, REV.ReviewDate, REV.Comments, REV.Rating
                    FROM
                        Employees AS EMP
                    JOIN
                        Departments AS DP ON EMP.DepartmentID = DP.DepartmentID
                    JOIN
                        Positions AS POST ON EMP.PositionID = POST.PositionID
                    LEFT JOIN
                        PerformanceReview AS REV ON EMP.EmployeeID = REV.EmployeeID;
            ')
        END;
END TRY

BEGIN CATCH
    PRINT CONCAT('ERROR OCCURED WHILE CREATING THE VIEW ''EmployeeManagementView'': ', ERROR_MESSAGE());
    PRINT CONCAT('ERROR CODE: ', CAST(@@ERROR AS NVARCHAR(10)));
END CATCH;
GO
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```

QUERY 09:

```
-- INSERT DATA INTO THE DEPARTMENTS TABLE
BEGIN TRY
    INSERT INTO Departments (DepartmentID, DepartmentName, DepartmentLocation, CreationDate)
        VALUES
            ('DEPT000001', 'HR', 'New York', '2023-01-01'),
            ('DEPT000002', 'Finance', 'London', '2023-01-02'),
            ('DEPT000003', 'IT', 'San Francisco', '2023-01-03'),
            ('DEPT000004', 'Sales', 'Chicago', '2023-01-04'),
            ('DEPT000005', 'Marketing', 'Los Angeles', '2023-01-05'),
            ('DEPT000006', 'Legal', 'Houston', '2023-01-06'),
            ('DEPT000007', 'R&D', 'Seattle', '2023-01-07'),
            ('DEPT000008', 'Operations', 'Boston', '2023-01-08'),
            ('DEPT000009', 'Customer Support', 'Miami', '2023-01-09'),
            ('DEPT000010', 'Administration', 'Dallas', '2023-01-10'),
            ('DEPT000011', 'Procurement', 'Denver', '2023-01-11'),
            ('DEPT000012', 'Logistics', 'Phoenix', '2023-01-12'),
            ('DEPT000013', 'Production', 'Philadelphia', '2023-01-13'),
            ('DEPT000014', 'Engineering', 'San Diego', '2023-01-14'),
            ('DEPT000015', 'Quality Assurance', 'Detroit', '2023-01-15'),
            ('DEPT000016', 'Product Management', 'Minneapolis', '2023-01-16'),
            ('DEPT000017', 'Compliance', 'Orlando', '2023-01-17'),
            ('DEPT000018', 'Investor Relations', 'San Jose', '2023-01-18'),
            ('DEPT000019', 'Public Relations', 'Charlotte', '2023-01-19'),
            ('DEPT000020', 'Facilities', 'Columbus', '2023-01-20');
END TRY

BEGIN CATCH
    PRINT CONCAT('ERROR OCCURED WHILE INSERTING RECORDS INTO THE ''DEPARTMENTS'' TABLE: ', ERROR_MESSAGE());
    PRINT CONCAT('ERROR CODE: ', CAST(@@ERROR AS NVARCHAR(10)));
END CATCH;
GO
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```

QUERY 10:

```
-- INSERT DATA INTO THE POSITIONS TABLE
BEGIN TRY
    INSERT INTO Positions (PositionID, PositionTitle, DepartmentID, MinSalary, MaxSalary, JobDescription)
        VALUES
            ('POST000001', 'HR Manager', 'DEPT000001', 50000, 70000, 'Manage HR department'),
            ('POST000002', 'Accountant', 'DEPT000002', 60000, 80000, 'Handle financial records'),
            ('POST000003', 'Software Engineer', 'DEPT000003', 80000, 120000, 'Develop software'),
            ('POST000004', 'Sales Manager', 'DEPT000004', 70000, 90000, 'Manage sales team'),
            ('POST000005', 'Marketing Specialist', 'DEPT000005', 60000, 85000, 'Develop marketing strategies'),
            ('POST000006', 'Legal Advisor', 'DEPT000006', 90000, 110000, 'Provide legal advice'),
            ('POST000007', 'Research Scientist', 'DEPT000007', 85000, 95000, 'Conduct research'),
            ('POST000008', 'Operations Manager', 'DEPT000008', 75000, 95000, 'Oversee operations'),
            ('POST000009', 'Customer Support Rep', 'DEPT000009', 40000, 60000, 'Support customers'),
            ('POST000010', 'Admin Assistant', 'DEPT000010', 35000, 50000, 'Administrative support'),
            ('POST000011', 'Procurement Officer', 'DEPT000011', 55000, 75000, 'Handle procurement'),
            ('POST000012', 'Logistics Coordinator', 'DEPT000012', 50000, 70000, 'Coordinate logistics'),
            ('POST000013', 'Production Supervisor', 'DEPT000013', 60000, 80000, 'Supervise production'),
            ('POST000014', 'Engineer', 'DEPT000014', 70000, 100000, 'Engineering tasks'),
            ('POST000015', 'QA Analyst', 'DEPT000015', 60000, 85000, 'Quality assurance'),
            ('POST000016', 'Product Manager', 'DEPT000016', 90000, 120000, 'Manage products'),
            ('POST000017', 'Compliance Officer', 'DEPT000017', 65000, 85000, 'Ensure compliance'),
            ('POST000018', 'IR Specialist', 'DEPT000018', 80000, 100000, 'Investor relations'),
            ('POST000019', 'PR Manager', 'DEPT000019', 75000, 95000, 'Public relations'),
            ('POST000020', 'Facilities Manager', 'DEPT000020', 60000, 80000, 'Manage facilities');
END TRY

BEGIN CATCH
    PRINT CONCAT('ERROR OCCURED WHILE INSERTING RECORDS INTO THE ''POSITIONS'' TABLE: ', ERROR_MESSAGE());
    PRINT CONCAT('ERROR CODE: ', CAST(@@ERROR AS NVARCHAR(10)));
END CATCH;
GO
-- FA22-BESE-0003 - FA22-BESE-0007 - FA22-BESE-0029
```

QUERY 11:

```
-- INSERT DATA INTO THE EMPLOYEES TABLE
BEGIN TRY
    INSERT INTO Employees (EmployeeID, FirstName, LastName, Email, PhoneNumber, HireDate, BaseSalary, Bonus, PayDate, DepartmentID, PositionID)
        VALUES
            ('EMP000001', 'John', 'Doe', 'john.doe@example.com', 1234567890, '2020-01-15', 50000, 5000, '2024-06-30', 'DEPT000001', 'POST000001'),
            ('EMP000002', 'Jane', 'Smith', 'jane.smith@example.com', 2345678901, '2019-03-20', 60000, 6000, '2024-06-30', 'DEPT000002', 'POST000002'),
            ('EMP000003', 'Robert', 'Brown', 'robert.brown@example.com', 3456789012, '2021-05-10', 55000, 5500, '2024-06-30', 'DEPT000003', 'POST000003'),
            ('EMP000004', 'Emily', 'Davis', 'emily.davis@example.com', 4567890123, '2018-07-25', 70000, 7000, '2024-06-30', 'DEPT000004', 'POST000004'),
            ('EMP000005', 'Michael', 'Wilson', 'michael.wilson@example.com', 5678901234, '2020-11-05', 50000, 4000, '2024-06-30', 'DEPT000004', 'POST000005'),
            ('EMP000006', 'Sarah', 'Miller', 'sarah.miller@example.com', 6789012345, '2019-08-15', 62000, 6200, '2024-06-30', 'DEPT000006', 'POST000006'),
            ('EMP000007', 'David', 'Martinez', 'david.martinez@example.com', 7890123456, '2022-03-18', 58000, 5800, '2024-06-30', 'DEPT000007', 'POST000007'),
            ('EMP000008', 'Laura', 'Anderson', 'laura.anderson@example.com', 8901234567, '2018-12-22', 51000, 5100, '2024-06-30', 'DEPT000008', 'POST000008'),
            ('EMP000009', 'James', 'Taylor', 'james.taylor@example.com', 9012345678, '2017-09-30', 56000, 5600, '2024-06-30', 'DEPT000009', 'POST000009'),
            ('EMP000010', 'Jessica', 'Thomas', 'jessica.thomas@example.com', 1123456789, '2016-06-01', 53000, 5300, '2024-06-30', 'DEPT000010', 'POST000010'),
            ('EMP000011', 'William', 'Jackson', 'william.jackson@example.com', 2234567890, '2021-11-11', 49000, 4900, '2024-06-30', 'DEPT000011', 'POST000011'),
            ('EMP000012', 'Olivia', 'White', 'olivia.white@example.com', 3345678901, '2020-10-19', 61000, 6100, '2024-06-30', 'DEPT000012', 'POST000012'),
            ('EMP000013', 'Joshua', 'Harris', 'joshua.harris@example.com', 4456789012, '2018-04-13', 54000, 5400, '2024-06-30', 'DEPT000013', 'POST000013'),
            ('EMP000014', 'Sophia', 'Clark', 'sophia.clark@example.com', 5567890123, '2019-07-29', 58000, 5800, '2024-06-30', 'DEPT000014', 'POST000014'),
            ('EMP000015', 'Daniel', 'Lewis', 'daniel.lewis@example.com', 6678901234, '2022-02-05', 50000, 5000, '2024-06-30', 'DEPT000015', 'POST000015'),
            ('EMP000016', 'Ava', 'Robinson', 'ava.robinson@example.com', 7789012345, '2021-06-23', 62000, 6200, '2024-06-30', 'DEPT000016', 'POST000016'),
            ('EMP000017', 'Matthew', 'Walker', 'matthew.walker@example.com', 8890123456, '2020-09-17', 57000, 5700, '2024-06-30', 'DEPT000017', 'POST000017'),
            ('EMP000018', 'Mia', 'Hall', 'mia.hall@example.com', 9901234567, '2019-05-21', 59000, 5900, '2024-06-30', 'DEPT000018', 'POST000018'),
            ('EMP000019', 'Christopher', 'Allen', 'christopher.allen@example.com', 1012345678, '2018-11-08', 52000, 5200, '2024-06-30', 'DEPT000019', 'POST000019'),
            ('EMP000020', 'Isabella', 'Young', 'isabella.young@example.com', 1123456789, '2017-08-03', 60000, 6000, '2024-06-30', 'DEPT000020', 'POST000020');
END TRY

BEGIN CATCH
    PRINT CONCAT('ERROR OCCURED WHILE INSERTING RECORDS INTO THE ''EMPLOYEES'' TABLE: ', ERROR_MESSAGE());
    PRINT CONCAT('ERROR CODE: ', CAST(@@ERROR AS NVARCHAR(10)));
END CATCH;
GO
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```

QUERY 12:

```
-- INSERT DATA INTO THE PERFORMANCE REVIEW TABLE
BEGIN TRY
    INSERT INTO PerformanceReview (ReviewID, EmployeeID, ReviewDate, Comments, Rating)
        VALUES
            ('REV000001', 'EMP000001', GETDATE(), 'Excellent performance.', 9),
            ('REV000002', 'EMP000002', GETDATE(), 'Good performance.', 7),
            ('REV000003', 'EMP000003', GETDATE(), 'Satisfactory performance.', 5),
            ('REV000004', 'EMP000004', GETDATE(), 'Needs improvement.', 4),
            ('REV000005', 'EMP000005', GETDATE(), 'Outstanding performance.', 1),
            ('REV000006', 'EMP000006', GETDATE(), 'Meets expectations.', 6),
            ('REV000007', 'EMP000007', GETDATE(), 'Below expectations.', 3),
            ('REV000008', 'EMP000008', GETDATE(), 'Exceeds expectations.', 8),
            ('REV000009', 'EMP000009', GETDATE(), 'Good teamwork.', 7),
            ('REV000010', 'EMP000010', GETDATE(), 'Poor performance.', 2),
            ('REV000011', 'EMP000011', GETDATE(), 'Excellent leadership.', 9),
            ('REV000012', 'EMP000012', GETDATE(), 'Satisfactory work.', 5),
            ('REV000013', 'EMP000013', GETDATE(), 'Room for growth.', 4),
            ('REV000014', 'EMP000014', GETDATE(), 'Highly dependable.', 8),
            ('REV000015', 'EMP000015', GETDATE(), 'Outstanding commitment.', 9),
            ('REV000016', 'EMP000016', GETDATE(), 'Average performance.', 6),
            ('REV000017', 'EMP000017', GETDATE(), 'Needs significant improvement.', 3),
            ('REV000018', 'EMP000018', GETDATE(), 'Consistently good.', 7),
            ('REV000019', 'EMP000019', GETDATE(), 'Excellent innovation.', 9),
            ('REV000020', 'EMP000020', GETDATE(), 'Underperformed.', 2);
END TRY

BEGIN CATCH
    PRINT CONCAT('ERROR OCCURED WHILE INSERTING RECORDS INTO THE ''PERFORMANCE REVIEW'' TABLE: ', ERROR_MESSAGE());
    PRINT CONCAT('ERROR CODE: ', CAST(@@ERROR AS NVARCHAR(10)));
END CATCH;
GO
-- FA22-BESE-0003 - FA22-BESE-0007 - FA22-BESE-0029
```

QUERY 13 & 14:

```
-- SHOW STORED DATA FROM THE DEPARTMENTS TABLE
BEGIN TRY
    SELECT * FROM Departments;
END TRY

BEGIN CATCH
    PRINT CONCAT('ERROR OCCURED WHILE SELECTING RECORDS FROM THE ''DEPARTMENTS'' TABLE: ', ERROR_MESSAGE());
    PRINT CONCAT('ERROR CODE: ', CAST(@@ERROR AS NVARCHAR(10)));
END CATCH;
GO
-- FA22-BESE-0003 - FA22-BESE-0007 - FA22-BESE-0029

-- SHOW STORED DATA FROM THE POSITIONS TABLE
BEGIN TRY
    SELECT * FROM Positions;
END TRY

BEGIN CATCH
    PRINT CONCAT('ERROR OCCURED WHILE SELECTING RECORDS FROM THE ''POSITIONS'' TABLE: ', ERROR_MESSAGE());
    PRINT CONCAT('ERROR CODE: ', CAST(@@ERROR AS NVARCHAR(10)));
END CATCH;
GO
-- FA22-BESE-0003 - FA22-BESE-0007 - FA22-BESE-0029
```

QUERY 15 & 16:

```
-- QUERY NUMBER 15:
-- SHOW STORED DATA FROM THE EMPLOYEES TABLE
BEGIN TRY
    SELECT * FROM Employees;
END TRY

BEGIN CATCH
    PRINT CONCAT('ERROR OCCURED WHILE SELECTING RECORDS FROM THE ''EMPLOYEES'' TABLE: ', ERROR_MESSAGE());
    PRINT CONCAT('ERROR CODE: ', CAST(@@ERROR AS NVARCHAR(10)));
END CATCH;
GO
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-- SHOW STORED DATA FROM THE PERFORMANCE REVIEWS TABLE
BEGIN TRY
    SELECT * FROM PerformanceReview;
END TRY

BEGIN CATCH
    PRINT CONCAT('ERROR OCCURED WHILE SELECTING RECORDS FROM THE ''PERFORMANCE REVIEW'' TABLE: ', ERROR_MESSAGE());
    PRINT CONCAT('ERROR CODE: ', CAST(@@ERROR AS NVARCHAR(10)));
END CATCH;
GO
-- FA22-BESE-0003 - FA22-BESE-0007 - FA22-BESE-0029
```

QUERY 17 & 18:

```
-- SHOW STORED DATA FROM THE SIGN_UP TABLE
BEGIN TRY
    SELECT * FROM Sign_Up;
END TRY

BEGIN CATCH
    PRINT CONCAT('ERROR OCCURED WHILE SELECTING RECORDS FROM THE ''SIGN-UP'' TABLE: ', ERROR_MESSAGE());
    PRINT CONCAT('ERROR CODE: ', CAST(@@ERROR AS NVARCHAR(10)));
END CATCH;
GO
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-- SHOW STORED DATA FROM THE EMPLOYEEMANAGEMENT VIEW
BEGIN TRY
    SELECT * FROM EmployeeManagementView;
END TRY

BEGIN CATCH
    PRINT CONCAT('ERROR OCCURED WHILE SELECTING RECORDS FROM THE ''EMPLOYEEMANAGEMENTVIEW'' TABLE: ', ERROR_MESSAGE());
    PRINT CONCAT('ERROR CODE: ', CAST(@@ERROR AS NVARCHAR(10)));
END CATCH;
GO
-- FA22-BESE-0003 - FA22-BESE-0007 - FA22-BESE-0029
```

QUERY 19 & 20:

```
-- UPDATING THE EMPLOYEES TABLE IN ORDER TO UPDATE THE VIEW AS MENTIONED IN THE CEP
BEGIN TRY
    UPDATE Employees
    SET FirstName = 'Kamran', LastName = 'Khurshid', Email = 'kamran.khurshid@example.com'
    WHERE EmployeeID = 'EMP000002'
END TRY

BEGIN CATCH
    PRINT CONCAT('ERROR OCCURED WHILE CREATING THE DATABASE: ', ERROR_MESSAGE());
    PRINT CONCAT('ERROR CODE: ', CAST(@@ERROR AS NVARCHAR(10)));
END CATCH;
GO
-- FA22-BESE-0003 - FA22-BESE-0007 - FA22-BESE-0029


-- SHOW STORED DATA FROM THE EMPLOYEEMANAGEMENT VIEW
BEGIN TRY
    SELECT * FROM EmployeeManagementView;
END TRY

BEGIN CATCH
    PRINT CONCAT('ERROR OCCURED WHILE SELECTING RECORDS FROM THE ''EMPLOYEEMANAGEMENTVIEW'' TABLE: ', ERROR_MESSAGE());
    PRINT CONCAT('ERROR CODE: ', CAST(@@ERROR AS NVARCHAR(10)));
END CATCH;
GO
-- FA22-BESE-0003 - FA22-BESE-0007 - FA22-BESE-0029
```

SQL RESULT SCREENSHOTS

RESULT 01:

```
(20 rows affected)

(20 rows affected)

(20 rows affected)

(20 rows affected)

(20 rows affected)

(20 rows affected)

(20 rows affected)

(20 rows affected)

(0 rows affected)

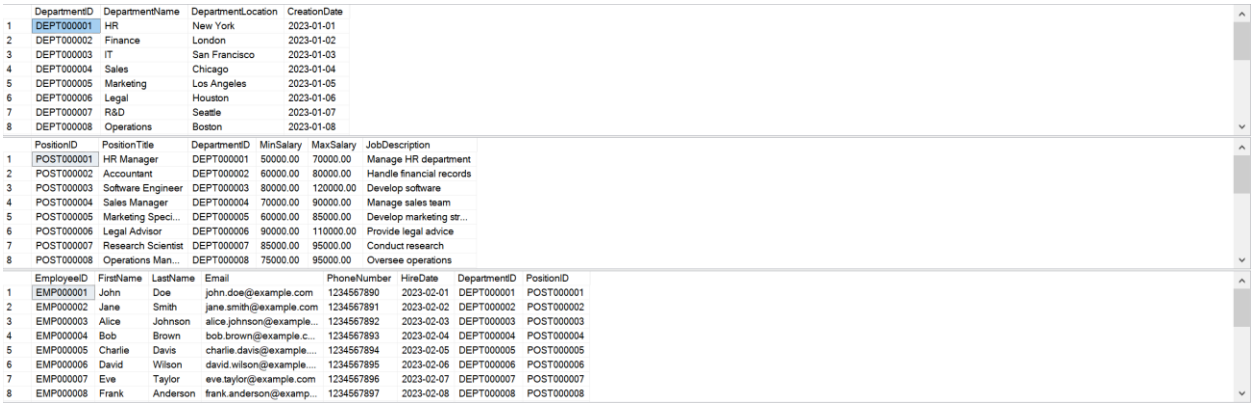
(20 rows affected)

(1 row affected)

(20 rows affected)

Completion time: 2024-06-04T20:18:47.5205985+05:00
```

RESULT 02:

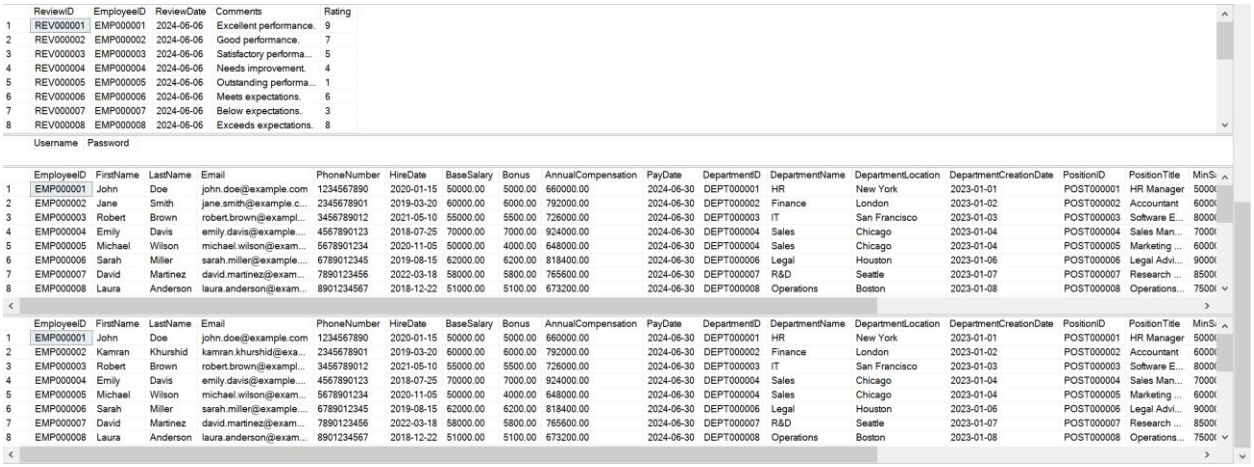


	DepartmentID	DepartmentName	DepartmentLocation	CreationDate
1	DEPT000001	HR	New York	2023-01-01
2	DEPT000002	Finance	London	2023-01-02
3	DEPT000003	IT	San Francisco	2023-01-03
4	DEPT000004	Sales	Chicago	2023-01-04
5	DEPT000005	Marketing	Los Angeles	2023-01-05
6	DEPT000006	Legal	Houston	2023-01-06
7	DEPT000007	R&D	Seattle	2023-01-07
8	DEPT000008	Operations	Boston	2023-01-08

	PositionID	PositionTitle	DepartmentID	MinSalary	MaxSalary	JobDescription
1	POST000001	HR Manager	DEPT000001	50000.00	70000.00	Manage HR department
2	POST000002	Accountant	DEPT000002	60000.00	80000.00	Handle financial records
3	POST000003	Software Engineer	DEPT000003	80000.00	120000.00	Develop software
4	POST000004	Sales Manager	DEPT000004	70000.00	90000.00	Manage sales team
5	POST000005	Marketing Speci...	DEPT000005	60000.00	85000.00	Develop marketing str...
6	POST000006	Legal Advisor	DEPT000006	90000.00	110000.00	Provide legal advice
7	POST000007	Research Scientist	DEPT000007	85000.00	95000.00	Conduct research
8	POST000008	Operations Man...	DEPT000008	75000.00	95000.00	Oversee operations

	EmployeeID	FirstName	LastName	Email	PhoneNumber	HireDate	DepartmentID	PositionID
1	EMP000001	John	Doe	john.doe@example.com	1234567890	2023-02-01	DEPT000001	POST000001
2	EMP000002	Jane	Smith	jane.smith@example.com	1234567891	2023-02-02	DEPT000002	POST000002
3	EMP000003	Alice	Johnson	alice.johnson@example...	1234567892	2023-02-03	DEPT000003	POST000003
4	EMP000004	Bob	Brown	bob.brown@example.c...	1234567893	2023-02-04	DEPT000004	POST000004
5	EMP000005	Charlie	Davis	charlie.davis@example....	1234567894	2023-02-05	DEPT000005	POST000005
6	EMP000006	David	Wilson	david.wilson@example....	1234567895	2023-02-06	DEPT000006	POST000006
7	EMP000007	Eve	Taylor	eve.taylor@example.com	1234567896	2023-02-07	DEPT000007	POST000007
8	EMP000008	Frank	Anderson	frank.anderson@examp...	1234567897	2023-02-08	DEPT000008	POST000008

RESULT 03:



	ReviewID	EmployeeID	ReviewDate	Comments	Rating
1	REV000001	EMP000001	2024-06-06	Excellent performance.	9
2	REV000002	EMP000002	2024-06-06	Good performance.	7
3	REV000003	EMP000003	2024-06-06	Satisfactory performa...	5
4	REV000004	EMP000004	2024-06-06	Needs improvement.	4
5	REV000005	EMP000005	2024-06-06	Outstanding performa...	1
6	REV000006	EMP000006	2024-06-06	Meets expectations.	6
7	REV000007	EMP000007	2024-06-06	Below expectations.	3
8	REV000008	EMP000008	2024-06-06	Exceeds expectations.	8

	Username	Password

	EmployeeID	FirstName	LastName	Email	PhoneNumber	HireDate	BaseSalary	Bonus	AnnualCompensation	PayDate	DepartmentID	DepartmentName	DepartmentLocation	DepartmentCreationDate	PositionID	PositionTitle	MinSa
1	EMP000001	John	Doe	john.doe@example.com	1234567890	2020-01-15	50000.00	5000.00	660000.00	2024-06-30	DEPT000001	HR	New York	2023-01-01	POST000001	HR Manager	5000
2	EMP000002	Jane	Smith	jane.smith@example.c...	2345678901	2019-03-20	60000.00	6000.00	792000.00	2024-06-30	DEPT000002	Finance	London	2023-01-02	POST000002	Accountant	6000
3	EMP000003	Robert	Brown	robert.brown@exampl...	3456789012	2021-05-10	55000.00	5500.00	726000.00	2024-06-30	DEPT000003	IT	San Francisco	2023-01-03	POST000003	Software E...	8000
4	EMP000004	Emily	Davis	emily.davis@example....	4567890123	2018-07-25	70000.00	7000.00	924000.00	2024-06-30	DEPT000004	Sales	Chicago	2023-01-04	POST000004	Sales Man...	7000
5	EMP000005	Michael	Wilson	michael.wilson@exam...	5678901234	2020-11-05	50000.00	4000.00	648000.00	2024-06-30	DEPT000004	Sales	Chicago	2023-01-04	POST000005	Marketing ...	6000
6	EMP000006	Sarah	Miller	sarah.miller@example....	6789012345	2019-08-15	62000.00	6200.00	818400.00	2024-06-30	DEPT000006	Legal	Houston	2023-01-06	POST000006	Legal Advi...	9000
7	EMP000007	David	Martinez	david.martinez@exam...	7890123456	2022-03-18	58000.00	5800.00	765600.00	2024-06-30	DEPT000007	R&D	Seattle	2023-01-07	POST000007	Research ...	8500
8	EMP000008	Laura	Anderson	laura.anderson@exam...	8901234567	2018-12-22	51000.00	5100.00	673200.00	2024-06-30	DEPT000008	Operations	Boston	2023-01-08	POST000008	Operations...	7500

	EmployeeID	FirstName	LastName	Email	PhoneNumber	HireDate	BaseSalary	Bonus	AnnualCompensation	PayDate	DepartmentID	DepartmentName	DepartmentLocation	DepartmentCreationDate	PositionID	PositionTitle	MinSa
1	EMP000001	John	Doe	john.doe@example.com	1234567890	2020-01-15	50000.00	5000.00	660000.00	2024-06-30	DEPT000001	HR	New York	2023-01-01	POST000001	HR Manager	5000
2	EMP000002	Kamran	Khurshid	kamran.khurshid@exa...	2345678901	2019-03-20	60000.00	6000.00	792000.00	2024-06-30	DEPT000002	Finance	London	2023-01-02	POST000002	Accountant	6000
3	EMP000003	Robert	Brown	robert.brown@exampl...	3456789012	2021-05-10	55000.00	5500.00	726000.00	2024-06-30	DEPT000003	IT	San Francisco	2023-01-03	POST000003	Software E...	8000
4	EMP000004	Emily	Davis	emily.davis@example....	4567890123	2018-07-25	70000.00	7000.00	924000.00	2024-06-30	DEPT000004	Sales	Chicago	2023-01-04	POST000004	Sales Man...	7000
5	EMP000005	Michael	Wilson	michael.wilson@exam...	5678901234	2020-11-05	50000.00	4000.00	648000.00	2024-06-30	DEPT000004	Sales	Chicago	2023-01-04	POST000005	Marketing ...	6000
6	EMP000006	Sarah	Miller	sarah.miller@example....	6789012345	2019-08-15	62000.00	6200.00	818400.00	2024-06-30	DEPT000006	Legal	Houston	2023-01-06	POST000006	Legal Advi...	9000
7	EMP000007	David	Martinez	david.martinez@exam...	7890123456	2022-03-18	58000.00	5800.00	765600.00	2024-06-30	DEPT000007	R&D	Seattle	2023-01-07	POST000007	Research ...	8500
8	EMP000008	Laura	Anderson	laura.anderson@exam...	8901234567	2018-12-22	51000.00	5100.00	673200.00	2024-06-30	DEPT000008	Operations	Boston	2023-01-08	POST000008	Operations...	7500

PYTHON IMPLEMENTATION

1) **Signup:** The signup feature allows new users to register by providing a username and password to the system. Passwords are converted to their hexadecimal representations for security concerns and then stored in the 'Sign_Up' table along with a unique username for each user, enhancing security.

2) **Login:** The login feature verifies user credentials by comparing the hash of the password entered at runtime with the password hashes that are already stored in the database. Successful verification of user credentials at the login page grants access to the employee management system (EMS), ensuring secure and reliable authentication.

3) **Add Department:** This feature enables the user to add new departments to the EMS. Administrators can create departments by providing the credentials requested by the system.

4) **Display Departments:** This feature retrieves the details of all departments from the database and displays them in the console, providing a list of departments existing within an organization and assisting users to view and manage organizational structure.

5) **Add Position:** Similar to adding departments, this feature allows the addition of new job positions to the 'positions' table.

6) **Display Positions:** This feature retrieves the details of all positions from the database and displays them in the console, providing a list of positions available within each department. It helps in understanding the job hierarchy and available roles.

7) **Add Employee:** This feature allows the addition of new employees to the 'Employees' table. The system will ask the user to enter employee details, such as employee ID, first name, last name, and email etc.

8) **Display Employees:** This feature retrieves the details of all employees from the database and displays them in the console, providing a list of employees working inside each department and position offering a comprehensive view of the workforce.

9) **Update Employee:** This functionality enables updating existing employee records. Changes to employee details are saved to the 'Employees' table, ensuring data accuracy.

10) **Remove Employee:** This feature allows the removal of employee records from the database. The specified employee is removed from the 'Employees' table.

11) **Add Performance Reviews:** This feature enables the user to add information regarding employee performance to the 'Performance Review' table by asking for details such as comments and performance rating etc.

12) **Display Performance Reviews:** This feature retrieves all records from the 'Performance Reviews' table and displays them in the console, providing a view of employees' performance.

PYTHON COSNOLE SCREENSHOTS

OUTPUT 01:

```
----------------------------------------
WELCOME TO EMPLOYEE MANAGEMENT SYSTEM!
0. Exit Employee Management System
1. SIGN-UP
2. LOG-IN
----------------------------------------
Enter Your Choice (0-1-2): 1
----------------------------------------
Enter Username: Faraz@1234
Enter Password: Faraz@1234
SIGN-UP SUCCESSFUL!
----------------------------------------
```

OUTPUT 02:

```
----------------------------------------
Enter Your Choice (0-1-2): 2
----------------------------------------
Enter Username: Faraz@1234
Enter Password: Faraz@1234
LOGIN SUCCESSFUL!
----------------------------------------
WELCOME TO EMPLOYEE MANAGEMENT SYSTEM!
0. Exit Employee Management System
1. Add Department
2. List Departments
3. Add Position
4. List Positions
5. Add Employee
6. List Employees
7. Update Employee's Details
8. Remove Employee
9. Add Salary
10. List Salaries
----------------------------------------
```

OUTPUT 03:

```
----------------------------------------
Enter Your Choice (0-1-10): 2
----------------------------------------
Department ID: DEPT000001
Department Name: HR
Department Location: New York
Creation Date: 2023-01-01

Department ID: DEPT000002
Department Name: Finance
Department Location: London
Creation Date: 2023-01-02

Department ID: DEPT000003
Department Name: IT
Department Location: San Francisco
Creation Date: 2023-01-03
```

OUTPUT 04:

```
----------------------------------------
Enter Your Choice (0-1-10): 4
----------------------------------------
Position ID: POST000001
Position Title: HR Manager
Department ID: DEPT000001
Minimum Salary: 50000.0000
Maximum Salary: 70000.0000
Job Description: Manage HR department

Position ID: POST000002
Position Title: Accountant
Department ID: DEPT000002
Minimum Salary: 60000.0000
Maximum Salary: 80000.0000
Job Description: Handle financial records

Position ID: POST000003
Position Title: Software Engineer
Department ID: DEPT000003
Minimum Salary: 80000.0000
Maximum Salary: 120000.0000
Job Description: Develop software
```

OUTPUT 05:

```
----------------------------------------
Enter Your Choice (0-1-10): 6
----------------------------------------
Employee ID: EMP000001
First Name: John
Last Name: Doe
Email: john.doe@example.com
Phone Number: 1234567890
Hire Date: 2020-01-15
Base Salary: 50000.0000
Bonus: 5000.0000
Annual Compensation: 660000.0000
Pay Date: 2024-06-30
Department ID: DEPT000001
Position ID: POST000001

Employee ID: EMP000002
First Name: Kamran
Last Name: Khurshid
Email: kamran.khurshid@example.com
Phone Number: 2345678901
Hire Date: 2019-03-20
Base Salary: 60000.0000
```

OUTPUT 06:

```
----------------------------------------
Enter Your Choice (0-1-10): 10
----------------------------------------
Review ID: REV000001
Employee ID: EMP000001
Review Date: 2024-06-06
Comments: Excellent performance.
Rating: 9

Review ID: REV000002
Employee ID: EMP000002
Review Date: 2024-06-06
Comments: Good performance.
Rating: 7

Review ID: REV000003
Employee ID: EMP000003
Review Date: 2024-06-06
Comments: Satisfactory performance.
Rating: 5
```

CONCLUSION

1) SQL SERVER: THE BEDROCK OF DATA INTEGRITY AND SECURITY

SQL Server serves as the rock-solid foundation of the EMS. Its robust data management capabilities ensure the accuracy and consistency of employee information. Features like data validation, access controls, and disaster recovery mechanisms safeguard sensitive employee data, fostering trust and compliance with regulations.

2) PYTHON: THE AGILE ENGINE FOR USER-FRIENDLY FUNCTIONALITY

Python, known for its readability and extensive libraries, acts as the agile engine driving the user-friendly interface of the EMS. This allows for the creation of intuitive dashboards, reports, and functionalities that streamline administrative tasks. Employees can access and update their information effortlessly, while HR personnel can leverage powerful analytics to gain insights into workforce trends.

3) A SYMPHONY OF STRENGTHS: A COMPREHENSIVE SOLUTION

The synergy between SQL Server and Python extends beyond individual strengths. Together, they enable the development of a comprehensive EMS solution. This solution can encompass a wide range of features, including:

- **Employee Onboarding and Offboarding:** Automate workflows for a smooth onboarding experience and efficient offboarding processes.
- **Performance Management:** Track employee performance through goal setting, progress monitoring, and feedback mechanisms.
- **Payroll Integration:** Integrate seamlessly with payroll systems for streamlined salary processing.
- **Reporting and Analytics:** Generate insightful reports on employee demographics, performance, and trends, empowering data-driven decision making.

4) A SCALABLE SOLUTION FOR FUTURE GROWTH

The chosen technology stack ensures the EMS is not just powerful but also scalable. SQL Server's ability to handle large datasets makes the system adaptable to growing organizations. Python's vast community and extensive libraries provide a wealth of resources for feature development and customization in the future.

In conclusion, the Employee Management System (EMS) implemented using SQL Server and Python offers a strategic partnership for organizations. It goes beyond efficiency, providing a comprehensive, secure, and user-friendly solution to manage your most valuable asset: your workforce. The system's comprehensive features support various administrative tasks, making it a valuable tool for any organization.